Performance improvements of commonly used optimizers in machine learning using **CUDA** and **OpenCL**

Lennart Hahner

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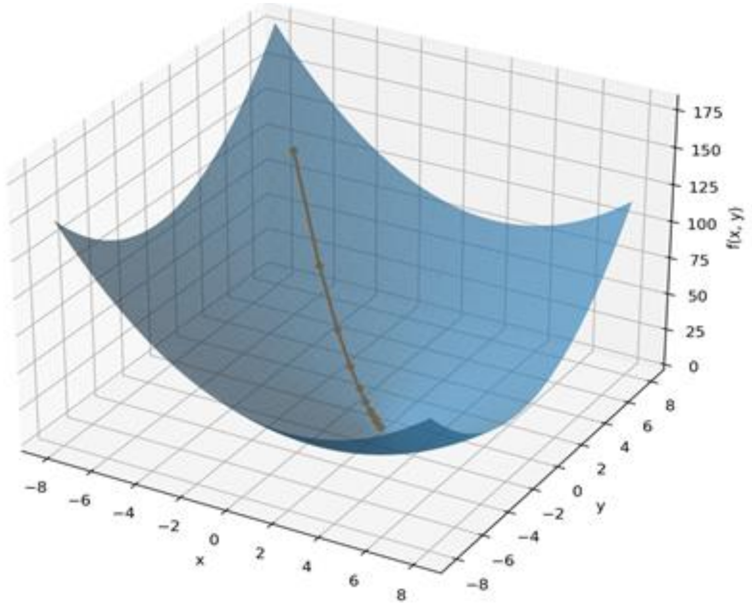
Motivation

The role of optimization in machine learning

Optimizers are the engines of machine learning models.

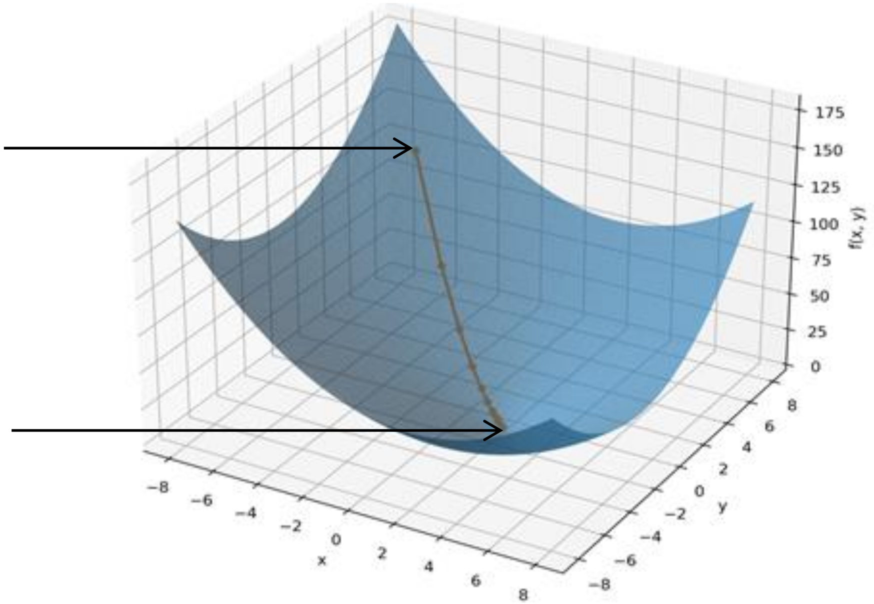


The role of optimization in machine learning

Train learnable weights towards a minimum of a loss function.

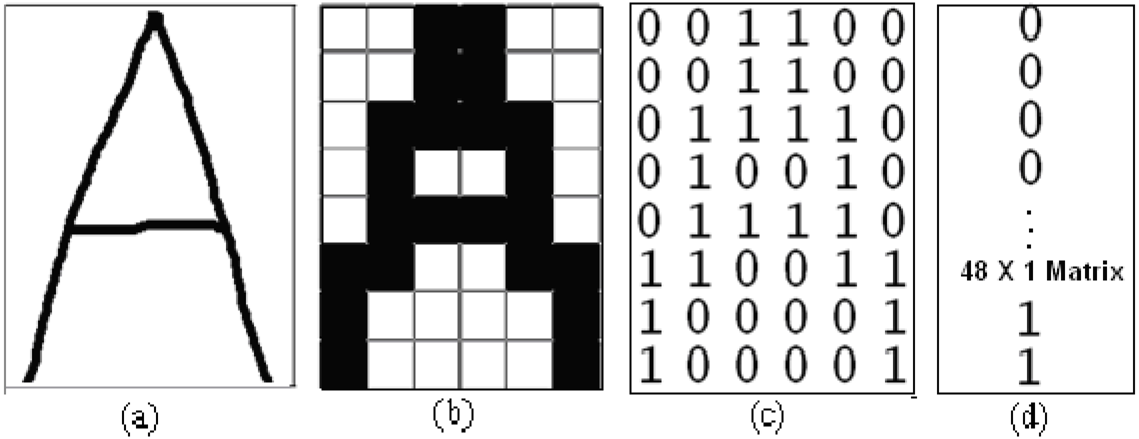
Start

Minima



The role of optimization in machine learning

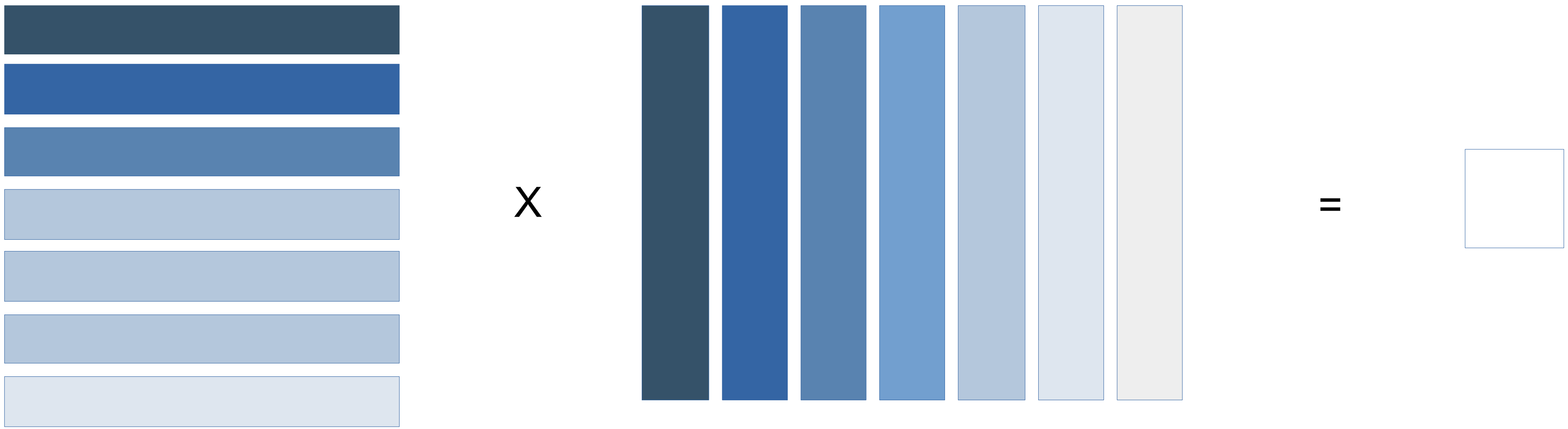
Most inputs i.e. images or text are represented as matrices...



[1] Figure 2 from Improving the Character Recognition Efficiency of Feed Forward BP Neural Network

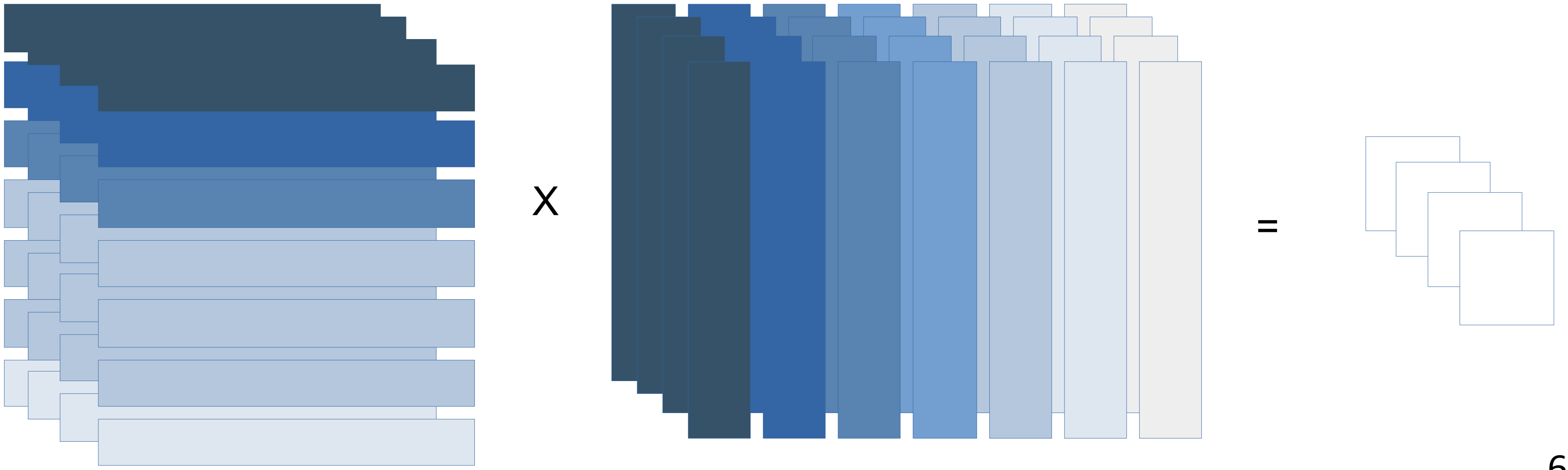
The role of hardware in machine learning

Thus we often deal with **matrix** or tensor operations during optimizations...



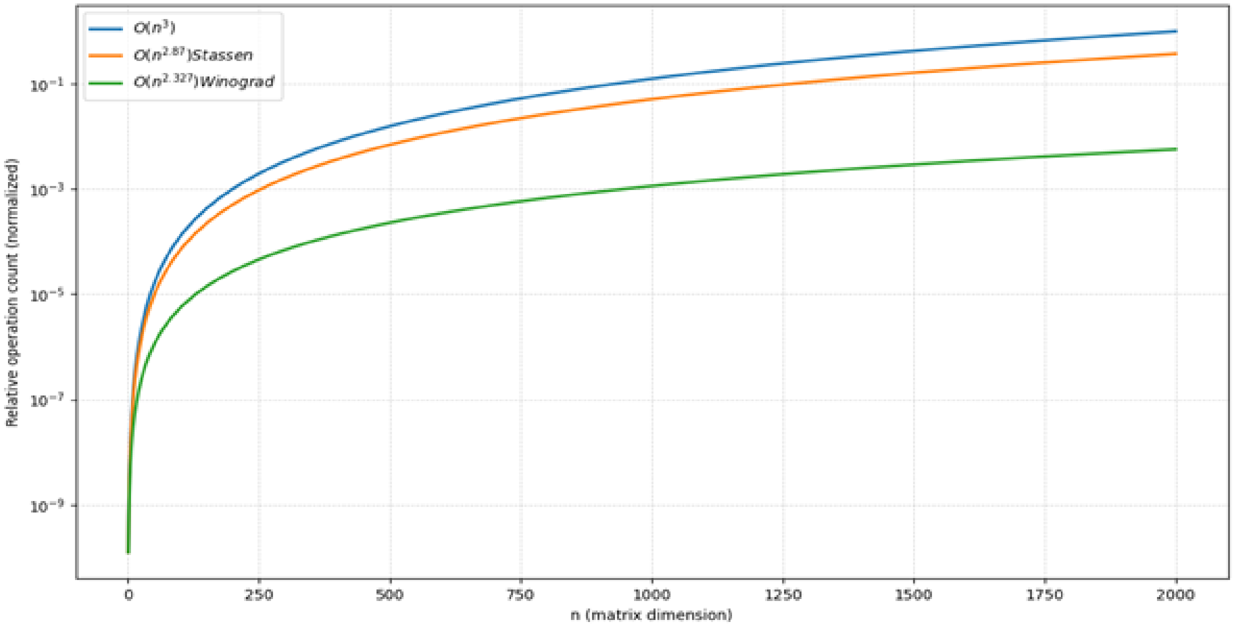
The role of hardware in machine learning

Thus we often deal with matrix or **tensor** operations during optimizations...



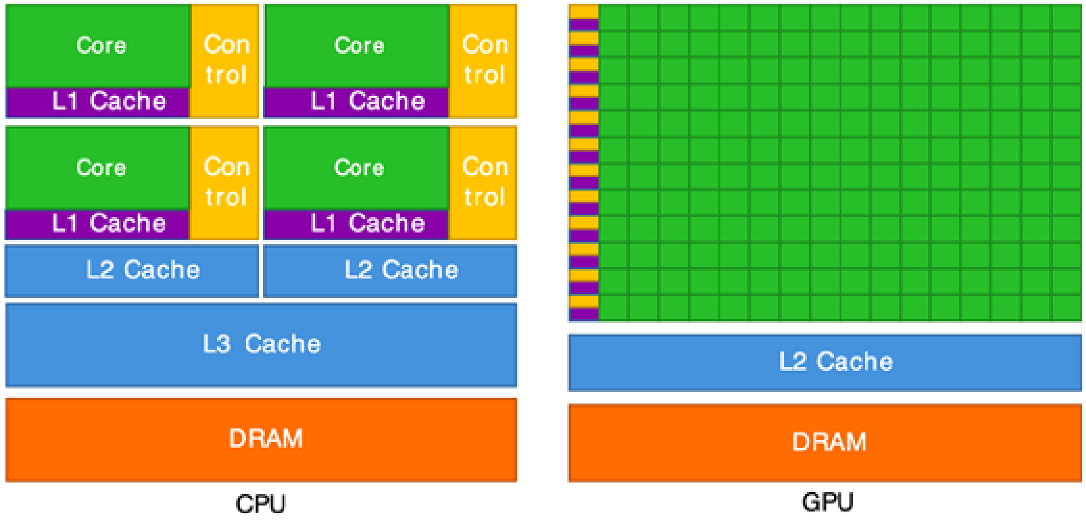
The role of hardware in machine learning

Matrix and Tensor operations are computationally expensive...



The role of hardware in machine learning

...exploiting hardware for parallelization



[2] Figure 2 - Review of GPU-based Monte Carlo simulation platforms for transmission and emission tomography in medicine.

The role of hardware in machine learning

...exploiting hardware for parallelization

Specification	**Intel Core i9 14900KF** [3]	**Nvidia GeForce RTX 5080** [4]
Cores	24	16384+
Max. Boost Clock (Ghz)	6	2.62
Architecture	x86	Blackwell
Core Paradigma	MIMD	SPMD

[3] Intel
[4] NVIDIA

The role of hardware in machine learning

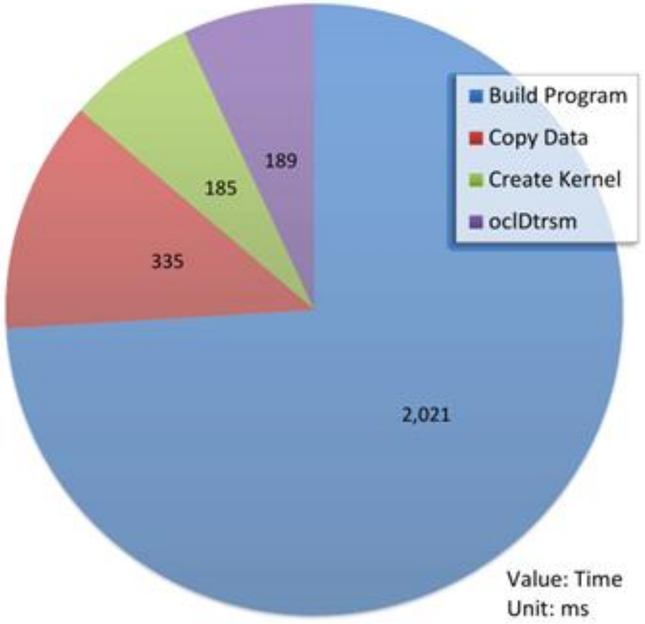
...how to exploit GPUs?

Expectations

Results of „From CUDA to OpenCL“ [2]

OpenCL enables code portability, but not performance

OpenCL GEMM program

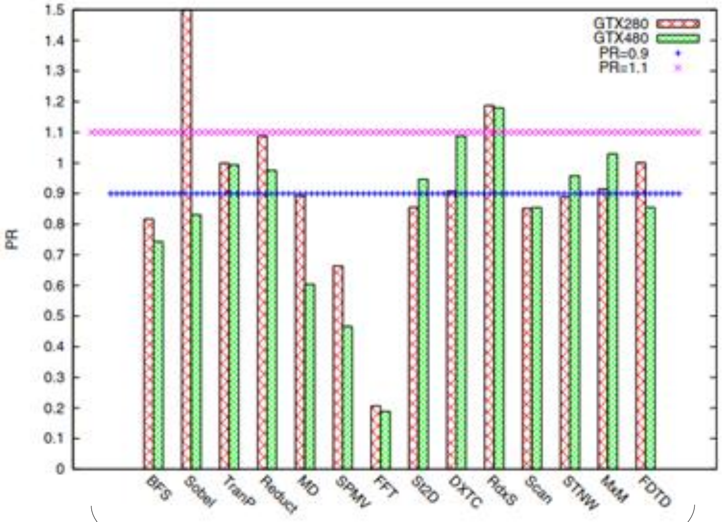


[3] „From CUDA to OpenCL: Towards a Performance-portable Solution for Multi-platform GPU Programming“ Peng D.

Results of „A comprehensive Performance comparision of CUDA and OpenCL“ [3]

CUDA is at most ~30% faster than OpenCL for most applications

$$PR = \frac{Perf(OpenCL)}{Perf(CUDA)}$$

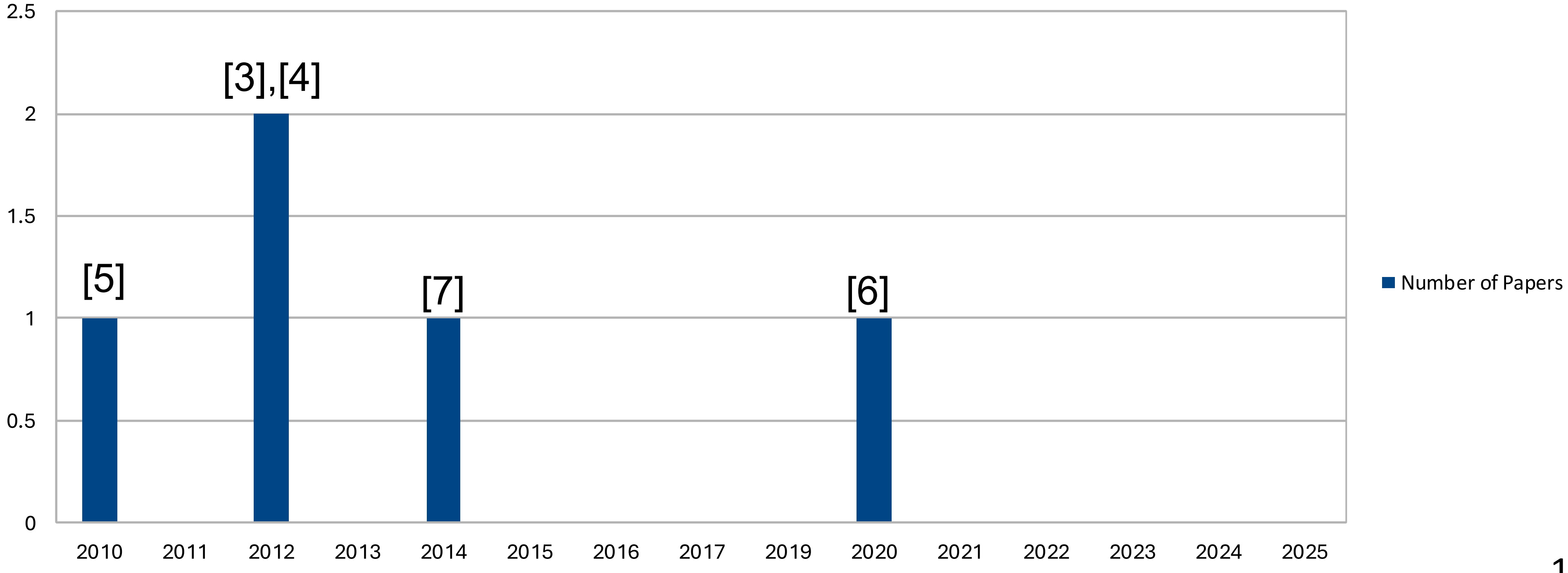


Workloads

[4] J. Fang, A. L. Varbanescu and H. Sips, "A Comprehensive Performance Comparison of CUDA and OpenCL,"

About more results

Most of the comparison results are more than 10 years old…



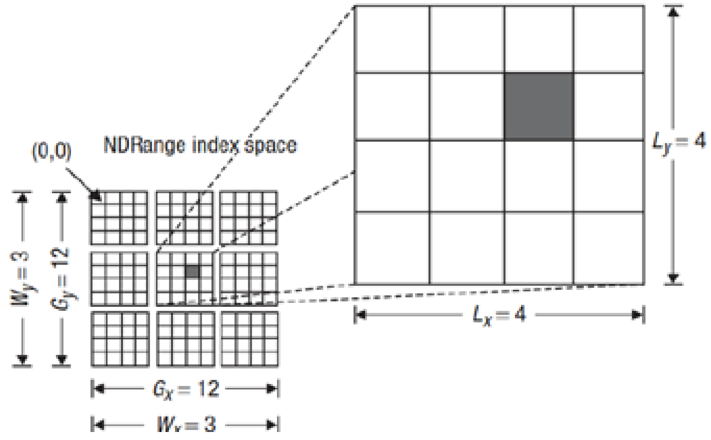
So what to expect from the comparison?

...CUDA might perform better, due to the longer compile/build time of OpenCL, but it’s worth revisiting...

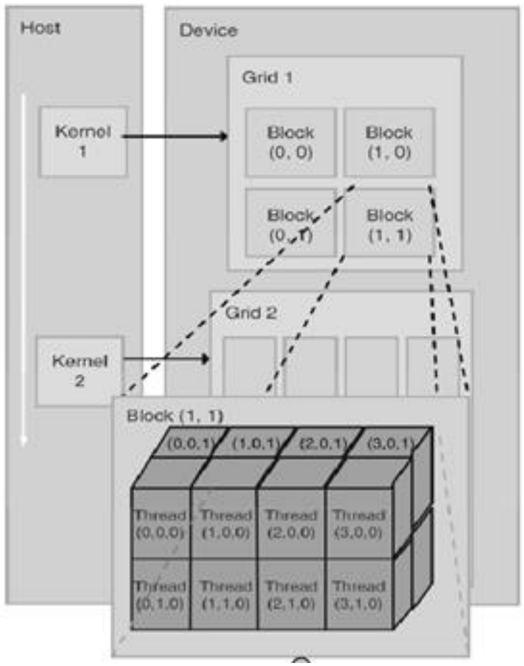
Conceptual comparison of CUDA and OpenCL

Abstraction differences

OpenCL™



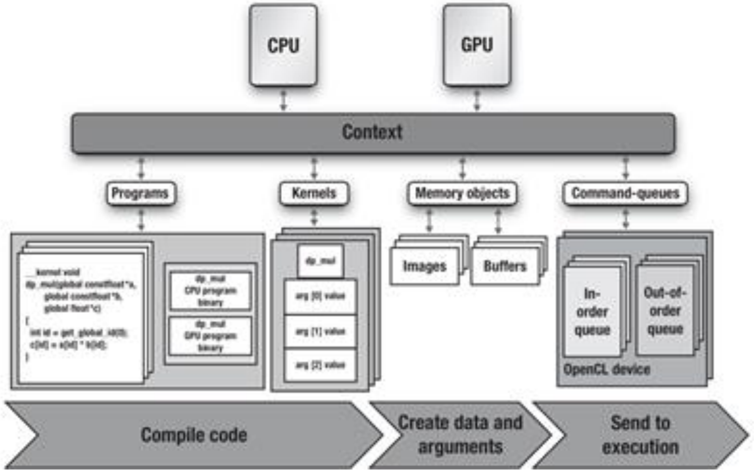
[8] Figure 1.7 in Aftab M. et al. "OpenCL Programming Guide"



TODO

[9] Figure 3.13 in Kirk D. et al. "Programming Massively Parallel Processors""

Architecture differences



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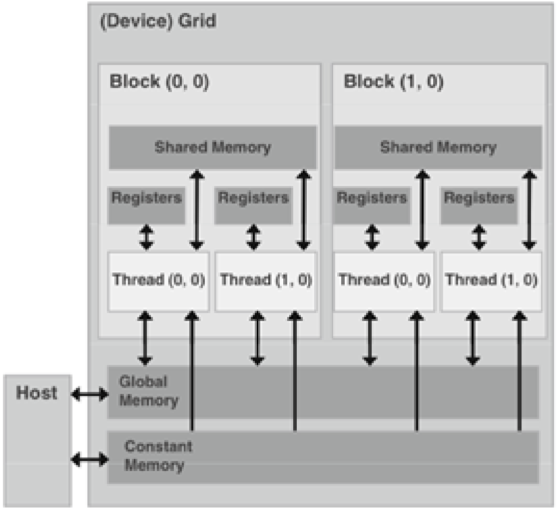
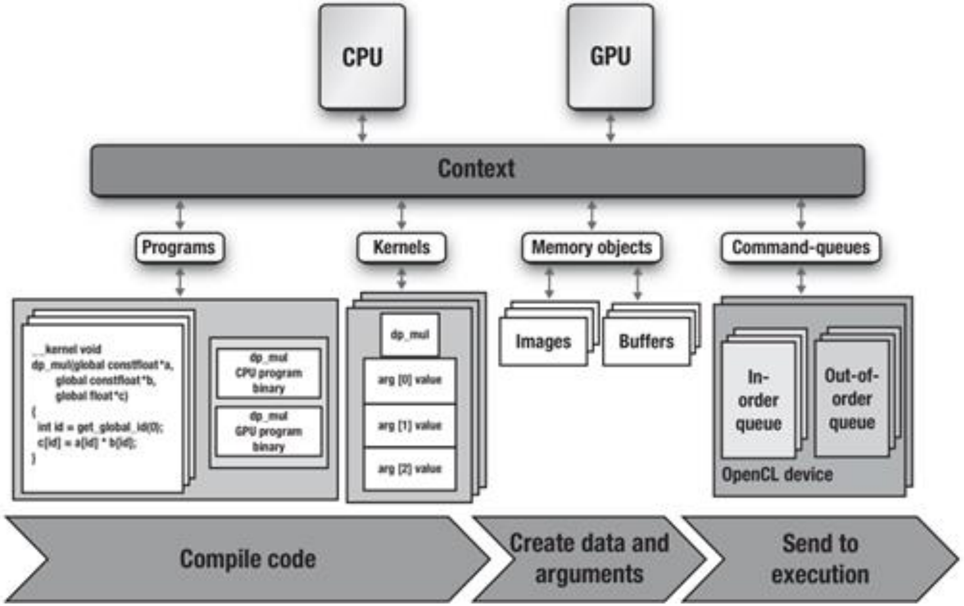


[9] Figure 3.13 in Kirk D. et al. “Programming Massively Parallel Processors””

Architecture differences

Unlike CUDA, OpenCL requires
an environmental setup...

1. Get `cl_device_id`
2. Create `cl_context`
3. Create `cl_command_queue`
4. Create `cl_program`
5. Define `cl_kernel`
6. Allocate `cl_mem`



[8] Figure 1.9 in Aftab M. et al. “OpenCL Programming Guide”

Architecture differences

NVCC enables easy integration of kernel functions...

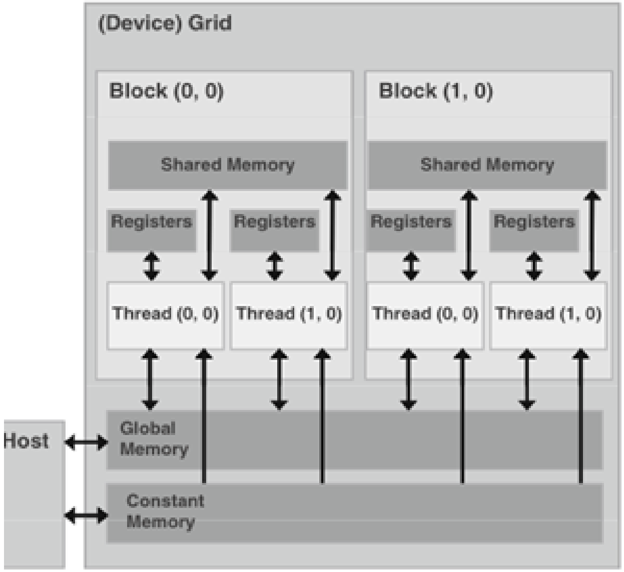
1. Allocate device memory `cudaMalloc(...)`

2. Copy to device `cudaMemcpy(...)`

3. Launch kernel `kernel«blocks, threads»(...)`

4. Copy from device `cudaMemcpy(...)`

5. Free `cudaFree(...)`



TODO

[9] Figure 3.13 in Kirk D. et al. “Programming Massively Parallel Processors””

Kernels in different languages

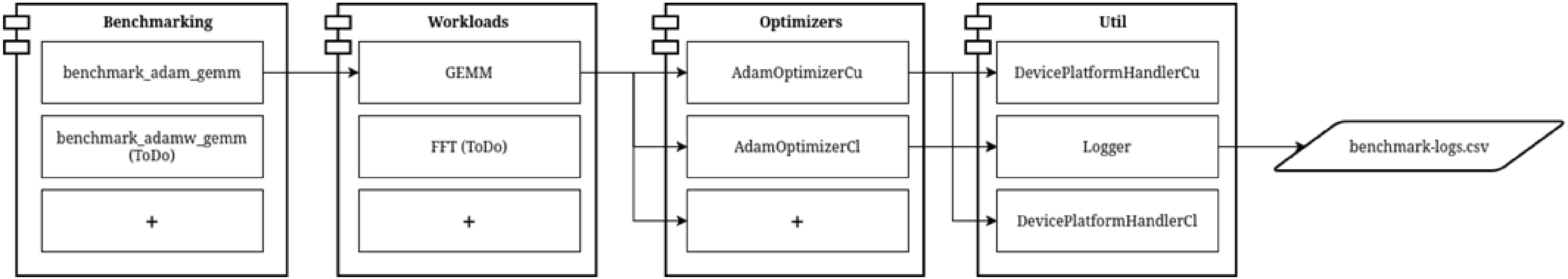
```
__global__ void adam_update_kernel_impl(
        float *__restrict__ param,
        const float *__restrict__ grad,
        float *__restrict__ m,
        float *__restrict__ v,
        size_t n,
        float lr, float beta1,
        float beta2,
        float bc1, float bc2, float eps)
    {
        ...
    }
```

```
__kernel void adam_update(
    __global float*       param,
    __global const float* grad,
    __global float*       m,
    __global float*       v,
    const float lr,
    const float beta1,
    const float beta2,
    const float eps,
    const float bc1,
    const float bc2,
    const int   n
){
    ...
}
```

The implementation

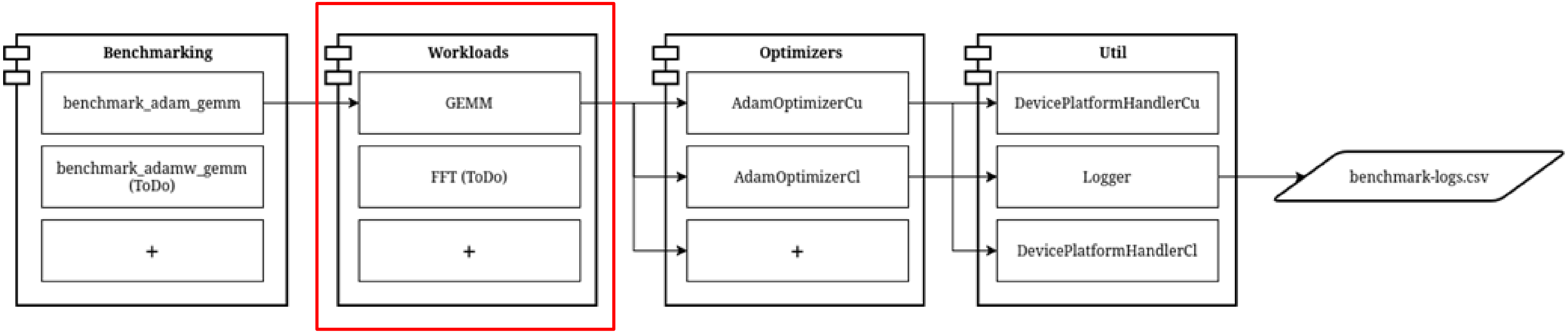
Architecture overview

Make it as extendable as possible for further permutations



Workload module

Definition and initialization of workloads...



Workload module

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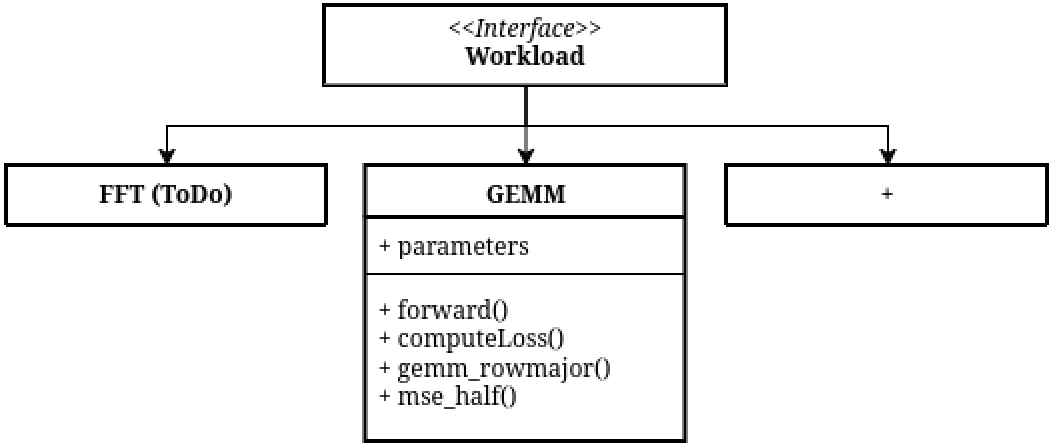
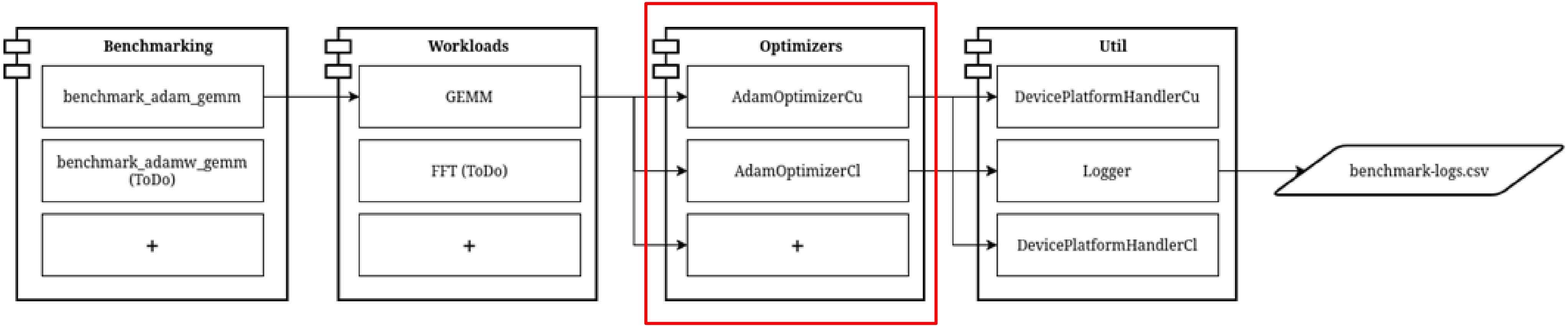
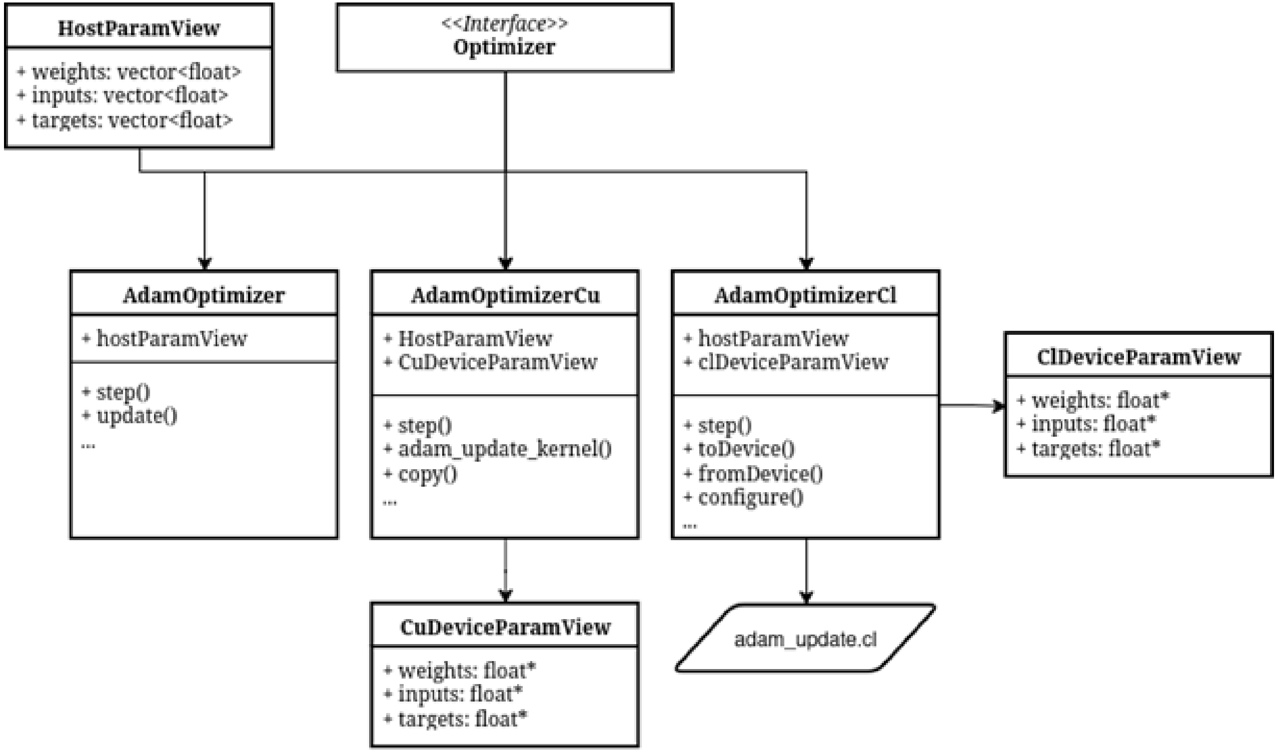


Optimizer module

The complete implementation of the optimizer in CUDA and OpenCL

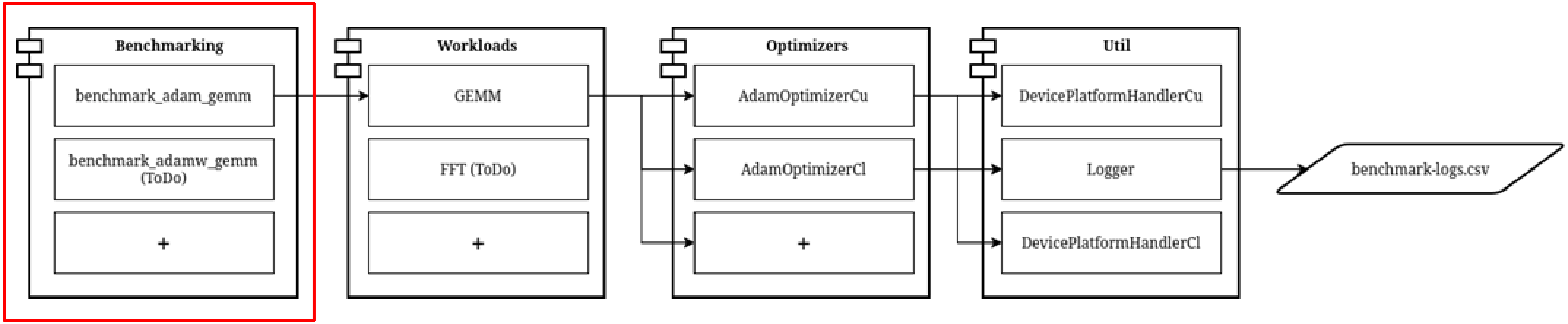


Optimizer module



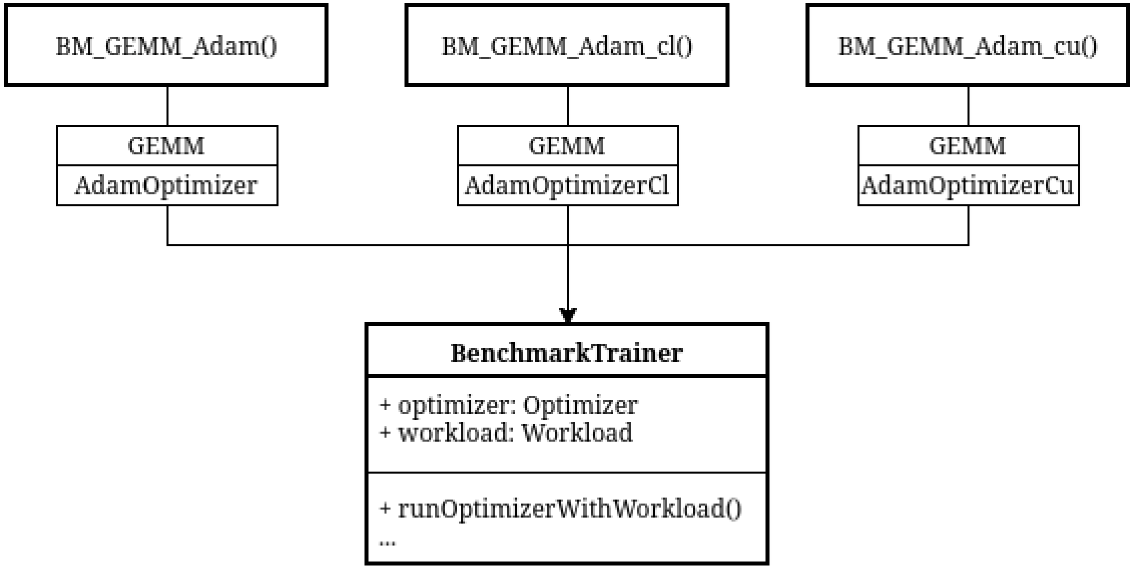
Benchmark module

The entry-point of the application

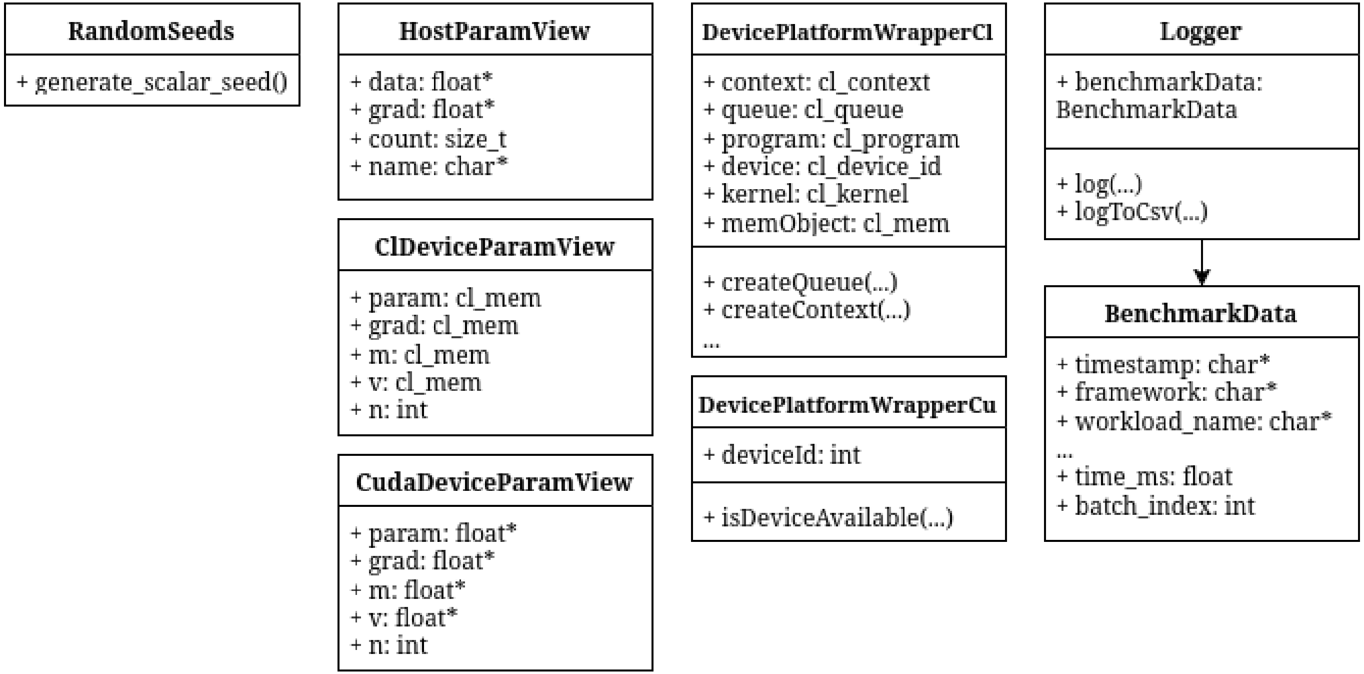


Benchmark module

The entry-point of the application



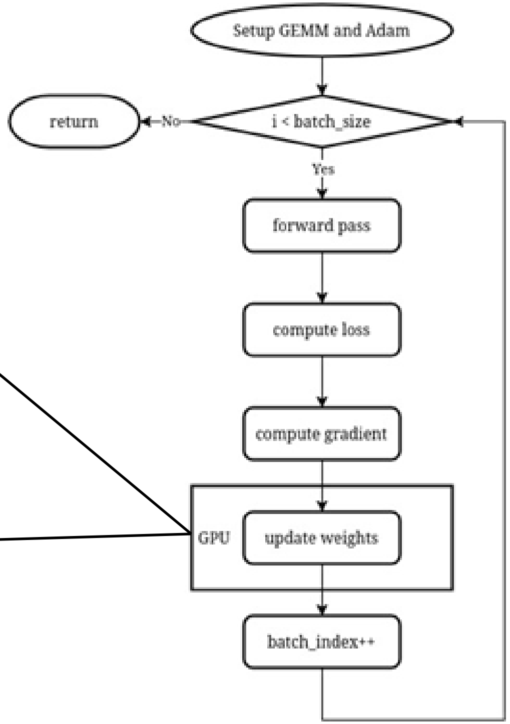
Util module


RandomSeeds
+ generate_scalar_seed()
HostParamView
+ data: float*
+ grad: float*
+ count: size_t
+ name: char*
ClDeviceParamView
+ param: cl_mem
+ grad: cl_mem
+ m: cl_mem
+ v: cl_mem
+ n: int
CudaDeviceParamView
+ param: float*
+ grad: float*
+ m: float*
+ v: float*
+ n: int
DevicePlatformWrapperCl
+ context: cl_context
+ queue: cl_queue
+ program: cl_program
+ device: cl_device_id
+ kernel: cl_kernel
+ memObject: cl_mem
+ createQueue(...)
+ createContext(...)
...
DevicePlatformWrapperCu
+ deviceId: int
+ isDeviceAvailable(...)
Logger
+ benchmarkData: BenchmarkData
+ log(...)
+ logToCsv(...)
BenchmarkData
+ timestamp: char*
+ framework: char*
+ workload_name: char*
...
+ time_ms: float
+ batch_index: int

Results

Benchmark environment

```
__global__ void adam_update_kernel_impl(
        float *__restrict__ param,
        const float *__restrict__ grad,
        float *__restrict__ m,
        float *__restrict__ v,
        size_t n,
        float lr, float beta1,
        float beta2,
        float bc1, float bc2, float eps)
    {
        ...
    }
```



Measured aspects in consideration

Metric	Measurement
Computation-Time	Measure start and end time difference of kernel execution.
Data-Transfer-time	Measure start and end time difference of host to device transfer.
Quality-of-solution	Compare the 10th batch run loss with expectiation and normalize
Build-time	Measure time to build kernel

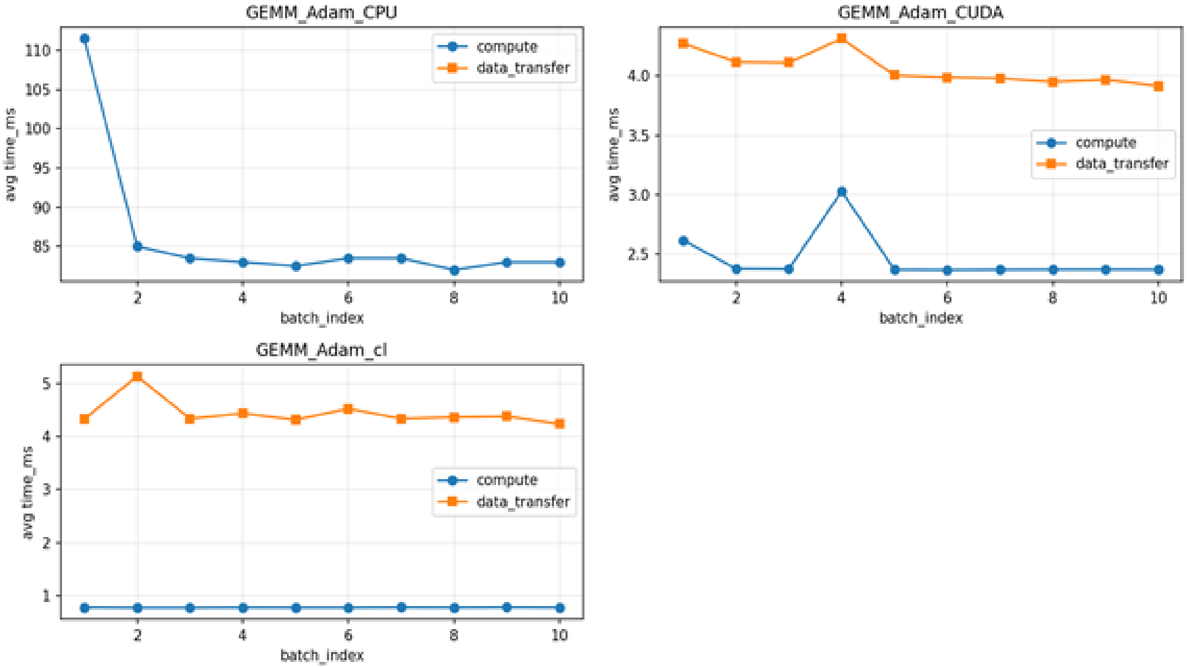
Benchmark environment

Running on…

A NVIDIA GeForce GTX 970
with an Intel i3-9100F and 16 GB RAM

Visuals of currently obtained measurements

Averaged computation time and
data transfer time on batch-index….



Conclusion so far

OpenCL and CUDA perform on average pretty similar…

...but some Remarks:

- ➢ Currently only tested on GTX 970 → **old platform**
- ➢ The number of results are **sparse**
- ➢ Some metrics are missing, e.g. **build-time**

Outlook

Some TODOs until end of march...

- ✗ Platform independence
- ✗ AdamW
- ✗ More measurements

Sources

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